

# INFINITE SIMPLE KAZHDAN GROUPS THAT ARE LIMITS OF FINITE SIMPLE GROUPS

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**ABSTRACT.** For every infinite minimal subshift  $X$ , the group  $G_X = \text{EL}_3(\text{LC}(X, \mathbb{F}_2) \rtimes \mathbb{Z})$  is an infinite, finitely generated, simple group with property (T) and a limit of finite simple groups  $\text{SL}_{3N}(\mathbb{F}_2)$  whose Cayley graphs form expanders. So  $G_X$  is locally embeddable into finite groups, sofic and hyperlinear, which answers the question of Brown and Ozawa whether an infinite simple Kazhdan group can be hyperlinear, and Pestov's sofic version of it. A finitely generated group is locally embeddable into finite groups if and only if it is a subgroup of an infinite finitely generated simple Kazhdan group that is a limit of finite simple groups whose Cayley graphs form expanders, and it also has solvable word problem if and only if the host can be chosen with solvable word problem. One such group contains every recursively presented finitely generated group with this property.

Can an infinite simple group with property (T) be hyperlinear? Brown asked this in 2001 as a question about embeddings into unitary groups of McDuff factors that embed in an ultrapower  $\mathcal{R}^\omega$  of the hyperfinite  $\text{II}_1$  factor [4, §11, Question 7]. Ozawa stated the hyperlinear form in 2003, noting that the full  $C^*$ -algebra of such a group cannot have the local lifting property [32, p. 527], and Pestov's Open question 9.1 adds the sofic form [33]. The groups below answer all three forms positively.

**Theorem 1.** *Let  $X \subseteq A^{\mathbb{Z}}$  be an infinite minimal subshift over a finite alphabet, with shift  $T$ , and let  $\text{LC}(X, \mathbb{F}_2)$  be the ring of locally constant functions  $X \rightarrow \mathbb{F}_2$ . Then*

$$G_X = \text{EL}_3(\text{LC}(X, \mathbb{F}_2) \rtimes_T \mathbb{Z})$$

*is an infinite, finitely generated, simple group with Kazhdan's property (T). It is the limit, in the space of marked groups, of finite simple groups  $\text{SL}_{3N}(\mathbb{F}_2)$  whose Cayley graphs, with respect to the images of a fixed generating set, form a family of expanders. So  $G_X$  is locally embeddable into finite groups (LEF), sofic and hyperlinear. The same holds for  $\text{EL}_n(\text{LC}(X, \mathbb{F}_2) \rtimes_T \mathbb{Z})$  for every  $n \geq 3$ , with  $\text{SL}_{nN}(\mathbb{F}_2)$  in place of  $\text{SL}_{3N}(\mathbb{F}_2)$ .*

A group is LEF [41] if every finite subset embeds injectively into a finite group, preserving the products that stay in that subset. The ring  $\text{LC}(X, \mathbb{F}_2) \rtimes_T \mathbb{Z}$  is finitely generated, so by the theorem of Ershov and Jaikin-Zapirain [11] the group  $G_X$  has Kazhdan's property (T) [22]. Periodic sequences approximating  $X$  give the ring finite models in the matrix algebras  $M_N(\mathbb{F}_2)$ , as in the proof by Grigorchuk and Medynets that topological full groups of minimal Cantor systems are LEF [16, Theorem 2.6]. So  $G_X$  is a limit of the groups  $\text{SL}_{3N}(\mathbb{F}_2)$ , which form expanders as in Kassabov's construction [21]. In the proof of simplicity, a nontrivial normal subgroup contains a nontrivial commutator lying in a copy of the finite simple group  $\text{GL}_d(\mathbb{F}_2)$  over a clopen tower, so it contains this group and with it an elementary matrix. For derived topological full groups, Matui showed by a similar argument that a nontrivial normal subgroup meets a

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simple union of alternating groups on towers [27, Lemma 3.4 and Theorem 4.9]. Simplicity also follows from Stepanov's theorem on the normal structure of  $\mathrm{GL}_n$  [37, Theorem 4.4]: the ring is simple [7, Corollary 4.6], and any  $p, q$  in it satisfy  $pr + qs = 0$  with  $(r, s) \neq 0$  by a dimension count on a tower over one small cylinder.

Hyperlinear groups were named by Rădulescu [35], and sofic groups were introduced by Gromov [18] and named by Weiss [42]. Thom constructed a finitely generated Kazhdan LEF group that is not residually finite and so lacks the factorization property [39], but his example is not simple. Pestov and Kwiatkowska then wrote that a hope of constructing groups that are not hyperlinear from property (T) remained, as it was unknown whether finitely generated simple Kazhdan groups can be hyperlinear [34].

The proof works for other actions. An action on a Cantor set is topologically free if no nontrivial element fixes a nonempty open set. We write  $e_{ij}(r) = I_n + rE_{ij}$ , with  $E_{ij}$  the matrix unit, for the elementary matrices.

**Theorem 2.** *Let a finitely generated group  $\Lambda$  act minimally and topologically freely on a Cantor set  $Z$ , and let  $R = \mathrm{LC}(Z, \mathbb{F}_2) \rtimes \Lambda$  be generated by a finite set  $S \ni 1$ . Suppose that there are maps  $\varphi_k: R \rightarrow M_{N_k}(\mathbb{F}_2)$  with  $\varphi_k(1) = I$  such that  $\varphi_k(S)$  generates the ring  $M_{N_k}(\mathbb{F}_2)$ , and for all  $r, s \in R$  we have  $\varphi_k(r + s) = \varphi_k(r) + \varphi_k(s)$  and  $\varphi_k(rs) = \varphi_k(r)\varphi_k(s)$  for large  $k$ . Then for every  $n \geq 3$ ,  $\mathrm{EL}_n(R)$  is an infinite, finitely generated, simple Kazhdan group. Marked by the  $e_{ij}(s)$  with  $s \in S$ , it is the limit of the finite simple groups  $\mathrm{SL}_{nN_k}(\mathbb{F}_2)$  marked by the  $e_{ij}(\varphi_k(s))$ , whose Cayley graphs form a family of expanders.*

Periodic approximations give such maps for  $\mathrm{LC}(X, \mathbb{F}_2) \rtimes_T \mathbb{Z}$ , which gives Theorem 1, and lamplighter actions show that the countable subgroups of the groups in Theorem 2 are exactly the countable LEF groups (Corollary 4).

## 1. PROOF OF THEOREM 2

We write the proof for  $n = 3$ ; for  $n \geq 3$  replace 3 by  $n$  throughout.

**The ring and property (T).** Write  $e_U$  for the indicator of a clopen set  $U \subseteq Z$ . The ring  $R = \bigoplus_{\xi \in \Lambda} \mathrm{LC}(Z, \mathbb{F}_2) u_\xi$  has the multiplication  $(f u_\xi)(f' u_\eta) = f(f' \circ \xi^{-1}) u_{\xi\eta}$ , so  $u_\xi e_U u_\xi^{-1} = e_{\xi U}$ . Fix a finite symmetric generating set of  $\Lambda$ , let  $|\xi|$  be the word length, and put  $B_m = \{\xi \in \Lambda : |\xi| \leq m\}$ . Let  $G = \mathrm{EL}_3(R)$  be the subgroup of  $\mathrm{GL}_3(R)$  generated by the  $e_{ij}(r)$  with  $i \neq j$  and  $r \in R$ . With  $[g, h] = ghg^{-1}h^{-1}$  and  $i, j, l$  distinct,

$$(1) \quad e_{ij}(r + s) = e_{ij}(r)e_{ij}(s), \quad [e_{il}(r), e_{lj}(s)] = e_{ij}(rs).$$

So the  $e_{ij}(s)$  with  $s \in S$  generate  $G$ . By Ershov and Jaikin-Zapirain [11, Theorem 1.1],  $\mathrm{EL}_n$  of every finitely generated associative unital ring has property (T) for  $n \geq 3$ . So  $G$  has property (T), and it is infinite because  $e_{12}(\mathrm{LC}(Z, \mathbb{F}_2))$  is infinite.

**Simplicity.** Let  $1 \neq K \trianglelefteq G$  and  $1 \neq g \in K$ , and let  $w \geq 0$  bound the word lengths of the  $\xi$  with  $u_\xi$  occurring in the entries of  $g$  and  $g^{-1}$ . Call a clopen set  $V$  small if  $V \cap \xi V = \emptyset$  for  $\xi \in B_{2w} \setminus \{e\}$  and every  $f \circ \xi$ , with  $\xi \in B_w$  and  $f$  a coefficient of an entry of  $g$  or  $g^{-1}$ , is constant on  $V$ . Each  $\xi \neq e$  fixes a closed set with empty interior, so the points moved by every  $\xi \in B_{2w} \setminus \{e\}$  are dense, and every nonempty clopen set contains a nonempty small one.

Some  $h = e_{ij}(e_V)$  with  $V$  small does not commute with  $g$ . Indeed,  $g$  commutes with  $e_{ij}(r)$  exactly when  $g_{pi}r = 0$  for  $p \neq i$ ,  $rg_{jq} = 0$  for  $q \neq j$ , and  $g_{ii}r = rg_{jj}$ . For  $c = \sum_\xi c_\xi u_\xi$  and small  $V$  we have  $ce_V = \sum_\xi c_\xi e_{\xi V} u_\xi$  and  $e_V c = \sum_\xi e_V c_\xi u_\xi$ . If  $c_\xi \neq 0$ , it equals 1 on a nonempty clopen set  $U$ , and a small set  $V \subseteq \xi^{-1}U$  gives  $ce_V \neq 0$ , while a small set  $V \subseteq U$  gives  $e_V c \neq 0$ . So if  $g$  commutes with every such  $h$ , then  $g$  is diagonal. In  $g_{ii}e_V = e_V g_{jj}$  the coefficients at  $u_\xi$

with  $\xi \neq e$  are supported in the disjoint sets  $\xi V$  and  $V$ , so they vanish, and the same argument shows that the corresponding coefficients of  $g_{ii}$  and  $g_{jj}$  vanish. The constant coefficients agree on every small  $V$ , so they agree everywhere. So  $g = cI_3$  with  $c \in \text{LC}(Z, \mathbb{F}_2)$ , and comparing constant coefficients in  $cc^{-1} = 1$  gives  $c = 1$  and  $g = 1$ .

Fix such  $h$  and  $V$ , and put  $d = 3|B_w|$ . For  $a, b \in B_w$  put  $\epsilon_{ab} = e_{aV}u_{ab^{-1}}$ . Since  $\epsilon_{ab}\epsilon_{a'b'} = e_{a(V \cap b^{-1}a'V)}u_{ab^{-1}a'b'^{-1}}$  and  $b^{-1}a' \in B_{2w}$ , we get  $\epsilon_{ab}\epsilon_{a'b'} = \delta_{ba'}\epsilon_{ab'}$ . As  $h \neq I_3$ ,  $V$  is nonempty, so the  $\epsilon_{ab}$  span a ring  $A_V \cong M_{|B_w|}(\mathbb{F}_2)$  with unit  $1_V = \sum_a \epsilon_{aa}$ , and  $y \mapsto (1 - 1_V)I_3 + y$  embeds  $\text{GL}_3(A_V) \cong \text{GL}_d(\mathbb{F}_2)$  in  $\text{GL}_3(R)$ . Its image  $H_V$  lies in  $G$ , since the transvections  $e_{pq}(\epsilon_{ab})$  with  $p \neq q$  generate it: the other transvections of  $\text{GL}_d(\mathbb{F}_2)$  are commutators of these.

Put  $x = [g, h] \in K \setminus \{1\}$ . If  $a, b \in B_w$  and  $f, f'$  are coefficients as above, then

$$f u_a e_V f' u_b = f e_{aV} (f' \circ a^{-1}) u_{ab} \in \{0, \epsilon_{a, b^{-1}}\},$$

since  $f \circ a$  and  $f'$  are constant on  $V$ . The entries of  $ghg^{-1} - I_3 = g e_V E_{ij} g^{-1}$  are sums of such products, and  $h^{-1} = h = I_3 + \epsilon_{ee} E_{ij}$ , so  $x - I_3 = (ghg^{-1} - h)h$  and  $x^{-1} - I_3 = h(ghg^{-1} - h)$  lie in  $M_3(A_V)$ . Then  $y = 1_V I_3 + (x - I_3)$  and  $y' = 1_V I_3 + (x^{-1} - I_3)$  satisfy  $yy' = y'y = 1_V I_3$ , so  $x = (1 - 1_V)I_3 + y \in H_V$ . As  $K \cap H_V$  is a nontrivial normal subgroup of  $H_V \cong \text{GL}_d(\mathbb{F}_2) = \text{PSL}_d(\mathbb{F}_2)$ , which is simple as  $d \geq 3$ , we get  $H_V \subseteq K$ , and  $e_{pq}(e_V) \in K$  for all  $p \neq q$ .

Finally,  $J = \{r \in R : e_{pq}(r) \in K \text{ for all } p \neq q\}$  is a two-sided ideal by (1), since  $e_{pq}(sr) = [e_{pl}(s), e_{lq}(r)]$  and  $e_{pq}(rs) = [e_{pl}(r), e_{lq}(s)]$  for  $l \notin \{p, q\}$ . It contains  $e_V$ , so it contains every  $e_{\xi V} = u_{\xi} e_V u_{\xi}^{-1}$ . By minimality and compactness finitely many of these sets cover  $Z$ , so  $1 = 1 - \prod_i (1 - e_{\xi_i V}) \in J$ . So  $J = R$  and  $K = G$ .

**Finite simple models.** Let  $F$  be the free algebra over  $\mathbb{F}_2$  on letters  $\tau_s, s \in S$ , and let  $\pi: F \rightarrow R$  and  $\rho_k: F \rightarrow M_{N_k}(\mathbb{F}_2)$  be the ring homomorphisms with  $\pi(\tau_s) = s$  and  $\rho_k(\tau_s) = \varphi_k(s)$ . Both are onto, so  $G$  and  $\text{EL}_3(M_{N_k}(\mathbb{F}_2))$  are quotients of  $\text{EL}_3(F)$ . Here  $\text{EL}_3(M_{N_k}(\mathbb{F}_2)) = \text{SL}_{3N_k}(\mathbb{F}_2)$ , as transvections generate  $\text{SL}_{3N_k}(\mathbb{F}_2)$  and those inside one block are commutators of those between blocks. The group  $\text{EL}_3(F)$  is generated by the  $e_{ij}(1)$  and  $e_{ij}(\tau_s)$ , and in both quotients  $e_{ij}(1)$  and  $e_{ij}(\tau_1)$  have the same image. A word of length  $\lambda$  or less in the  $e_{ij}(\tau_s)$  has entries of degree at most  $\lambda$  in  $F$ , and  $\varphi_k \circ \pi = \rho_k$  on the finitely many such entries for large  $k$ . So a word that is trivial in  $G$  is trivial in  $\text{SL}_{3N_k}(\mathbb{F}_2)$  for large  $k$ . Conversely, for a nonprincipal ultrafilter  $\omega$  on  $\mathbb{N}$  the  $\varphi_k$  define a unital ring homomorphism from  $R$  to the algebraic ultraproduct  $\prod_{\omega} M_{N_k}(\mathbb{F}_2)$ . It maps  $G$  to  $\prod_{\omega} \text{SL}_{3N_k}(\mathbb{F}_2)$ , and its kernel on  $G$  is a normal subgroup that does not contain  $e_{12}(1)$ , so it is trivial. As this holds for every  $\omega$ , each nontrivial word is nontrivial in  $\text{SL}_{3N_k}(\mathbb{F}_2)$  for all but finitely many  $k$ . So these finite simple groups converge to  $G$  in the space of marked groups [15, 6],  $G$  is LEF, and  $N_k \rightarrow \infty$  as  $G$  is infinite. Since  $\text{EL}_3(F)$  has property (T) [11, Theorem 1.1], its finite quotients form a family of expanders [26, 21].  $\square$

## 2. PROOF OF THEOREM 1

Let  $\mathbb{Z}$  act on  $X$  by powers of  $T$ , where  $(Tx)_t = x_{t+1}$ , and write  $u = u_1$ , so  $ufu^{-1} = f \circ T^{-1}$ . Minimality and infiniteness imply that  $X$  has no isolated points and  $T$  has no periodic points, so  $X$  is a Cantor set and the action is topologically free. The ring  $R = \text{LC}(X, \mathbb{F}_2) \rtimes_{\mathbb{Z}} \mathbb{Z}$  is generated by  $S = \{1, u, u^{-1}\} \cup \{e_a : a \in A\}$ , where  $e_a = e_{\{x: x_0=a\}}$ , since products of the translates  $u^j e_a u^{-j}$  are the cylinder indicators, which span  $\text{LC}(X, \mathbb{F}_2)$ . We write  $\tau_{\pm}$  and  $\tau_a$  for the letters of  $u^{\pm 1}$  and  $e_a$ .

Fix  $x \in X$  and  $\ell \geq 0$ . By minimality the forward orbit of  $x$  is dense, so every word of  $X$  occurs in  $x_{[0, \infty)}$ , and  $x_{[-\ell, \ell]}$  recurs at arbitrarily large positions. So there is  $m$  with  $x_{[m-\ell, m+\ell]} = x_{[-\ell, \ell]}$  such that every word of length  $2\ell + 1$  of  $X$  occurs in  $x_{[0, m]}$ . The  $m$ -periodic sequence  $y_{\ell}$  that agrees with  $x$  on  $[0, m)$  then agrees with  $x$  on  $[-\ell, m + \ell]$ . Every window of length  $2\ell + 1$  of  $y_{\ell}$  is

a translate of one centered in  $[0, m)$ , so  $y_\ell$  has the same words of length  $2\ell + 1$  as  $X$ . Let  $N_\ell$  be its least period, and on  $\mathbb{F}_2^{\mathbb{Z}/N_\ell\mathbb{Z}}$  let  $P\delta_t = \delta_{t+1}$ . For  $f$  depending only on coordinates in  $[-\ell, \ell]$  let  $D_\ell(f)\delta_t = f(T^t y_\ell)\delta_t$ , where  $f(T^t y_\ell)$  is the value of  $f$  at any point of  $X$  that agrees with  $T^t y_\ell$  on  $[-\ell, \ell]$ . Put  $\varphi_\ell(\sum_j f_j u^j) = \sum_j D_\ell(f_j)P^j$  when every  $f_j$  is of this kind, and  $\varphi_\ell = 0$  otherwise. Since  $PD_\ell(f)P^{-1} = D_\ell(f \circ T^{-1})$  whenever  $f$  depends only on coordinates in  $[-\ell + 1, \ell - 1]$ , for fixed  $r, s \in R$  the identities  $\varphi_\ell(r + s) = \varphi_\ell(r) + \varphi_\ell(s)$  and  $\varphi_\ell(rs) = \varphi_\ell(r)\varphi_\ell(s)$  hold for large  $\ell$ , and  $\varphi_\ell(1) = I$ . The shifts  $T^t y_\ell$ ,  $0 \leq t < N_\ell$ , are distinct, so products of the  $P^i D_\ell(e_a)P^{-i}$  give every diagonal matrix unit, and with  $P$  every matrix unit. So  $\varphi_\ell(S)$  generates  $M_{N_\ell}(\mathbb{F}_2)$ , and Theorem 2 applies.  $\square$

**Brown's formulation.** Let  $G$  be a group as in Theorem 2. LEF groups are sofic [33, Example 4.5], and sofic groups are hyperlinear [10, Theorem 2]. So  $L(G)$  embeds in  $\mathcal{R}^\omega$  [24, 35], [32, Proposition 7.1]. Since  $G$  is infinite and simple, its nontrivial conjugacy classes are infinite and  $G$  is not residually finite. So  $L(G)$  is a  $\text{II}_1$  factor, and  $L(G) \bar{\otimes} \mathcal{R}$  is a McDuff factor [28] that embeds in  $\mathcal{R}^\omega$  and whose unitary group contains  $G$ , as in Brown's question.

### 3. LEF GROUPS

Kionke and Schesler asked which groups embed in finitely generated simple amenable groups [23]. For LEF groups the next corollary answers the analogous question, with a Kazhdan host. Its second statement is the analogue for LEF groups of the theorem of Boone and Higman that a finitely generated group has solvable word problem if and only if it embeds in a simple subgroup of a finitely presented group [3], in Thompson's form, where the simple group is finitely generated and has solvable word problem [40].

**Corollary 3.** *A finitely generated group is LEF if and only if it is a subgroup of an infinite, finitely generated, simple group with property (T) that is a limit of finite simple groups  $\text{SL}_N(\mathbb{F}_2)$  in the space of marked groups, whose Cayley graphs with respect to the images of a fixed generating set form a family of expanders. It is LEF with solvable word problem if and only if it is a subgroup of such a group with solvable word problem.*

**Corollary 4.** *Every countable LEF group is a subgroup of such a group. A set of finitely generated groups lies in one such group if and only if its members are LEF and represent countably many isomorphism classes. So one such group contains every recursively presented finitely generated LEF group, for instance every finitely presented residually finite group and every  $G_Y$  with  $L(Y)$  recursive (Corollary 5).*

Simple hosts with only one of property (T) and LEF were known. Without property (T), Kionke and Schesler proved that every finitely generated residually finite group embeds in a finitely generated simple LEF group [23, Theorem 1.2], so with Wilson's theorem [43] one simple LEF group contains every finitely presented residually finite group. Without LEF, every countable group embeds in a finitely generated simple group [14, 36], and even in a simple quotient of a torsion-free hyperbolic Kazhdan group [8, Theorem 1.5], so one simple Kazhdan group contains Higman's universal finitely presented group [20].

*Proof.* Limits of finite groups in the space of marked groups are LEF, and so are their subgroups. Conversely, let  $\Gamma = \{\gamma_1, \gamma_2, \dots\}$  be a countable LEF group, replaced by  $\Gamma \times \mathbb{Z}$  so that it is infinite. There are finite groups  $Q_k$  and maps  $\mu_k: \Gamma \rightarrow Q_k$  that are injective on  $\{1, \gamma_1, \dots, \gamma_k\}$  and preserve the products that stay in this set. On  $Q_k \times \mathbb{Z}/m_k$ , with  $m_k = |Q_k|$ , let  $t_k$  add 1 to the second coordinate, and let  $f_{jk}$  multiply the first coordinate of  $(x, i)$  from the left by  $\mu_k(\gamma_j)^i$ , which is well defined as  $|Q_k|$  divides  $m_k$ . Then  $[f_{jk}, t_k]$  multiplies the first coordinate by  $\mu_k(\gamma_j)$ ,

as in the embedding of B. H. Neumann and H. Neumann [30]. On  $Q_k \times \mathbb{Z}/m_k \times \mathbb{Z}/4^{k+1}$  let  $b_k$  add 1 to the last coordinate, and let  $a_k$  act on the level  $i$  by  $f_{jk}$  if  $i = 2^{2j-1}$  and by  $t_k$  if  $i = 4^j$ , for  $j \leq k$ , and trivially on the other levels. Then  $b_k^{-n} a_k b_k^n$  acts on the level  $i$  as  $a_k$  acts on the level  $i + n$ . As distinct pairs of powers of 2 have distinct differences, all below a quarter of the modulus, the word  $w_j = [b^{-2^{2j-1}} a b^{2^{2j-1}}, b^{-4^j} a b^{4^j}]$  evaluated at  $(a_k, b_k)$  acts by  $[f_{jk}, t_k]$  on the level 0 and trivially on the other levels, for  $j \leq k$ . In the algebraic ultraproduct over  $\omega$  of the symmetric groups of these sets, sending  $\gamma$  to the class of the permutations multiplying the first coordinate on the level 0 by  $\mu_k(\gamma)$  and acting trivially elsewhere is a homomorphism, injective as  $\mu_k(\gamma) \neq 1$  for  $\gamma \neq 1$  and large  $k$ . It sends  $\gamma_j$  to  $w_j(a, b)$ , where  $a = (a_k)_k$  and  $b = (b_k)_k$ , so  $\Gamma \leq [\Delta, \Delta]$  for the group  $\Delta$  generated by  $a$  and  $b$ . Finitely many products and inequalities in the ultraproduct hold in some coordinate, so  $\Delta$  is infinite, finitely generated and LEF.

Let  $\Delta$  act on  $\Omega = \mathbb{F}_2^\Delta$  by  $(\delta x)(\eta) = x(\delta^{-1}\eta)$ , and let  $\Lambda$  be the group of maps  $x \mapsto \delta x + c$  with  $\delta \in \Delta$  and  $c$  finitely supported. It is generated by  $\Delta$  and the map  $z$  adding 1 at  $e$ , and its orbits contain the dense sets  $x + \bigoplus_\Delta \mathbb{F}_2$ . If  $(\delta, c) \neq (e, 0)$ , the fixed points of  $x \mapsto \delta x + c$  satisfy  $x(\eta) = x(\delta^{-1}\eta) + c(\eta)$  for all  $\eta$ . In every cylinder over a finite set  $W$  this fails at some point, for  $\eta \in \text{supp } c$  if  $\delta = e$  and for  $\eta \notin W \cup \delta W \cup \text{supp } c$  if  $\delta \neq e$ . So the action is minimal and topologically free. The ring  $R_\Delta = \text{LC}(\Omega, \mathbb{F}_2) \rtimes \Lambda$  is generated by 1, the units  $u_s^{\pm 1}$  for the generators  $s$  of  $\Lambda$ , and the indicator  $e_U$  of  $U = \{x : x(e) = 1\}$ .

Let  $\sigma_n : \Delta \rightarrow Q'_n$  be finite models of  $\Delta$  that are injective on the ball of radius  $n$  and preserve the products that stay in it, with  $\sigma_n(e) = e$  and  $Q'_n$  generated by the images of the generators. Let  $x \mapsto \delta x + c$  act on  $Y_n = \mathbb{F}_2^{Q'_n}$  by  $y \mapsto \sigma_n(\delta)y + \sum_{\eta \in \text{supp } c} 1_{\sigma_n(\eta)}$ , where  $(qy)(q') = y(q^{-1}q')$ , and attach to  $y$  the point  $y \circ \sigma_n$  of  $\Omega$ . For fixed elements of  $\Lambda$  and large  $n$ , these permutations multiply as in  $\Lambda$  and, uniformly in  $y$ , move the attached points near  $e$  as  $\Lambda$  moves the points of  $\Omega$ . Let  $D_n(f)$  be the diagonal matrix on  $\mathbb{F}_2^{Y_n}$  with entry  $f(y \circ \sigma_n)$  at  $y$ , let  $P_\xi$  be the permutation matrix of  $\xi \in \Lambda$ , and put  $\varphi_n(\sum_\xi f_\xi u_\xi) = \sum_\xi D_n(f_\xi) P_\xi$ . For fixed  $f$  and  $\xi$  and large  $n$ ,

$$P_\xi D_n(f) P_\xi^{-1} = D_n(f \circ \xi^{-1}),$$

so for fixed  $r, s \in R_\Delta$  and large  $n$ ,  $\varphi_n$  is additive and multiplicative on  $r, s$ , and  $\varphi_n(1) = I$ . The conjugates of  $D_n(e_U)$  by products of the  $P_s$  are the coordinate functions of  $Y_n$ , and the conjugates of  $P_z$  are the translations by the  $1_q$ , so the images of the generators generate  $M_{2|Q'_n|}(\mathbb{F}_2)$ . By Theorem 2,  $G_\Delta = \text{EL}_3(R_\Delta)$  is an infinite finitely generated simple Kazhdan group, and the limit of finite simple groups  $\text{SL}_{3N}(\mathbb{F}_2)$  whose Cayley graphs form expanders.

Finally,  $\xi \mapsto \text{diag}(u_\xi, 1, 1)$  is an injective homomorphism  $\Lambda \rightarrow \text{GL}_3(R_\Delta)$ . It maps  $[\Lambda, \Lambda]$  into  $G_\Delta$ , since over  $\mathbb{F}_2$  every unit  $v$  satisfies  $\text{diag}(v, v^{-1}, 1) = e_{12}(v) e_{21}(v^{-1}) e_{12}(v) e_{12}(1) e_{21}(1) e_{12}(1)$ , and the product of these matrices for  $v = v_1, v_2$  and  $(v_2 v_1)^{-1}$  is  $\text{diag}(v_1 v_2 v_1^{-1} v_2^{-1}, 1, 1)$ . So  $G_\Delta$  contains  $\Gamma \leq [\Delta, \Delta] \leq [\Lambda, \Lambda]$ . This proves the first statements of Corollaries 3 and 4. For the second statement of Corollary 4, apply the first to the restricted direct sum of representatives of countably many isomorphism classes of finitely generated LEF groups, which is a countable LEF group, as finite subsets lie in finite products. Conversely, a countable group has countably many finitely generated subgroups, and there are countably many recursive presentations.

Now let  $\Gamma$  be generated by a finite set  $E$ , LEF and with solvable word problem. A half-line form of the Neumann–Neumann construction, used by Darbinyan to preserve solvable word problems [9], keeps the word problem solvable. On  $\Gamma \times \mathbb{Z}$ , whose second coordinate is the level, let  $\beta$  raise the level by 1, and let  $h_\gamma$  multiply the first coordinate from the left by  $\gamma$  on the levels  $m \geq 0$ . Then  $[\beta, h_\gamma]$  multiplies the level 0 by  $\gamma^{-1}$ , so  $\Delta = \langle \beta, h_\gamma : \gamma \in E \rangle$  is infinite and  $\Gamma \leq [\Delta, \Delta]$ . An element of  $\Delta$  of length at most  $r$  is  $f\beta^\ell$  with  $|\ell| \leq r$ , where  $f$  multiplies the level  $m$  from the left by an element  $f(m)$  of length at most  $r$ , with  $f(m) = 1$  for  $m < -r$  and  $f(m) = f(r)$  for  $m > r$ . Let  $\psi$  map the ball of radius  $2r$  of  $\Gamma$  injectively

into a finite group  $Q$ , preserving the products that stay in it, and let  $\theta$ , with  $\theta(v)_{-3r} = 1$  and  $\theta(v)_m = v_{m-1}$  otherwise, be the isomorphism from  $\{v \in Q^{[-3r, 3r]} : v_{3r-1} = v_{3r}\}$  onto  $\{v \in Q^{[-3r, 3r]} : v_{-3r} = 1\}$ . Transversals give a bijection  $\kappa$  of  $Q^{[-3r, 3r]}$  with  $\kappa(vx) = \theta(v)\kappa(x)$  for  $v$  in the domain of  $\theta$ . Let each  $v \in Q^{[-3r, 3r]}$  act on  $Q^{[-3r, 3r]} \times \mathbb{Z}/5r$  by left multiplication on the first coordinate, and put  $\zeta(x, i) = (\kappa(x), i + 1)$ , so that  $\zeta v \zeta^{-1} = \theta(v)$  on the domain of  $\theta$ . As  $f$  is trivial below  $-r$  and constant from  $r$  on,  $\zeta^\ell(\psi(f(m)))_m \zeta^{-\ell} = (\psi(f(m - \ell)))_m$  for  $|\ell| \leq r$ . So  $f\beta^\ell \mapsto (\psi(f(m)))_{|m| \leq 3r} \zeta^\ell$  preserves the products that stay in the ball of radius  $r$ , and it is injective there, as  $5r > 2r$  and  $f$  is determined by its values for  $|m| \leq 3r$ . So  $\Delta$  is LEF, and the argument above applies to it.

From a word in the  $\beta^{\pm 1}$  and  $h_\gamma^{\pm 1}$  one computes the exponent sum  $\ell$  of  $\beta$ , the levels where  $f$  changes, namely the exponent sums of  $\beta$  before the letters  $h_\gamma^{\pm 1}$ , and the values of  $f$  there as words in  $E$ . It is trivial if and only if  $\ell = 0$  and these values are trivial, so  $\Delta$  has solvable word problem. An element  $x \mapsto \delta x + c$  of  $\Lambda$  is given by a word for  $\delta$  and words for the lamps of  $c$ , so the word problem of  $\Delta$  decides equality in  $\Lambda$  and computes the values of  $c$ . A word in the generators of  $G_\Delta$  multiplies out to a matrix with entries  $\sum_\xi p_\xi u_\xi$ , where each  $p_\xi$  is a polynomial over  $\mathbb{F}_2$  in finitely many coordinates  $x(\delta)$ , as  $u_\xi e_U u_\xi^{-1}$  is the indicator of  $\{x : x(\delta) = 1 + c(\delta)\}$  for  $\xi : x \mapsto \delta x + c$ . After collecting equal  $\xi$  and equal coordinates, the word is trivial if and only if every coefficient of its difference from  $I_3$  vanishes at every assignment of its finitely many variables, as  $\Omega$  is the full shift. So  $G_\Delta$  has solvable word problem. Conversely, finitely generated subgroups of groups with solvable word problem have solvable word problem, and LEF passes to subgroups. This proves the second statement of Corollary 3.  $\square$

#### 4. WORD PROBLEMS

**Corollary 5.** *The word problem of  $G_X$  has the Turing degree of the language  $L(X)$  of  $X$ . In particular  $G_X$  has solvable word problem if and only if  $L(X)$  is recursive, as for the Fibonacci subshift  $X_{(3-\sqrt{5})/2}$  below. Every Turing degree occurs, and there are continuum many groups  $G_X$ , none of which is isomorphic to a subgroup of another.*

*Proof.* Multiplying out a word in the generators in  $\text{LC}(A^\mathbb{Z}, \mathbb{F}_2) \rtimes \mathbb{Z}$ , which maps onto  $R$ , gives a matrix with entries  $\sum_j f_j u^j$ , each  $f_j$  given by a table on the words of some length, using  $uf = (f \circ T^{-1})u$ . The word is trivial in  $G_X$  if and only if the tables of its difference from  $I_3$  vanish on  $L(X)$ , so  $L(X)$  computes the word problem. Conversely, by (1) a word for  $e_{12}(\prod_{t < m} u^{-t} e_{v_t} u^t)$  is computable from  $v = v_0 \cdots v_{m-1}$ , and it is trivial if and only if  $v \notin L(X)$ .

For irrational  $\alpha \in (0, 1)$  let  $X_\alpha$  be the infinite minimal Sturmian subshift, the closure of the codings  $c(\theta)$ ,  $\theta \in [0, 1)$ , with  $c(\theta)_t = \lfloor \theta + (t+1)\alpha \rfloor - \lfloor \theta + t\alpha \rfloor$ , which is 1 exactly for  $\theta$  in the arc  $[-(t+1)\alpha, -t\alpha]$  modulo 1 [29, 19], [25, Chapter 2]. Passing to the closure adds no words, so the words of length  $m$  of  $X_\alpha$  are the constant values of  $c(\theta)_{[0, m]}$  on the arcs between the distinct points  $-j\alpha \bmod 1$ ,  $0 \leq j \leq m$ , and  $\alpha$  computes  $L(X_\alpha)$ . Such a word has  $\lfloor \theta + m\alpha \rfloor$  ones, within 1 of  $m\alpha$ , so  $L(X_\alpha)$  computes  $\alpha$ . Every  $D \subseteq \mathbb{N}$  has the degree of  $[0; 1 + \chi_D(0), 1 + \chi_D(1), \dots]$ , since each digit is decided by one strict comparison with a rational computed from earlier digits. If a finitely generated group  $H$  embeds in a finitely generated group  $G$ , a word for  $H$  becomes a word for  $G$  by substituting for its generators, so the word problem of  $H$  reduces to that of  $G$ . By a construction of Sacks, the Turing degrees contain an antichain of size continuum [31, Chapter V], and the groups  $G_{X_\alpha}$  with  $\alpha$  of these degrees are as claimed.  $\square$

For derived topological full groups, Grigorchuk and Medynets proved that the word problem is decidable if and only if  $L(X)$  is recursive [17, Theorem 1.1(3)].

## 5. QUESTIONS

Is there an infinite finitely presented simple group with property (T) that is sofic, or at least hyperlinear? Infinite finitely presented simple Kazhdan groups exist [5], while  $G_X$  is not finitely presented: a finitely presented LEF group is residually finite [38, 41], and  $G_X$  is not. A sofic example would not be LEF, and it would also answer Open problem 6.1 of Alekseev and Thom [1], which asks for finitely presented sofic groups with property (T) that are not residually finite. Is every sofic infinite simple Kazhdan group LEF?

Is every finitely generated sofic group a subgroup of an infinite finitely generated simple Kazhdan sofic group, as in Corollary 3?

If  $(X, T)$  is topologically conjugate to  $(Y, T')$  or to  $(Y, T'^{-1})$ , then  $G_X \cong G_Y$ . For topological full groups the converse holds [13, 2]. Does  $G_X \cong G_Y$  imply that  $X$  and  $Y$  are flip conjugate, or at least strongly orbit equivalent [12]? Does  $L(G_X) \cong L(G_Y)$  imply  $G_X \cong G_Y$ ?

## ORIGIN AND AUTHORSHIP

Claude (Anthropic) found the construction and original proofs under the author's direction on September 12, 2026, and wrote the manuscript. The author is responsible for the final manuscript. Proof records are in the [project repository](#).

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